

	JERUSALEM COLLEGE OF ENGINEERING (An Autonomous Institution, Affiliated to Anna University, Chennai)
	QUESTION BANK

Subject code : JIT2321

Subject Name: Digital Principles and computer Organization

Year / Semester : II/ III

Regulation : 2023

Branch : CSE/CSBS/IT/CS/AIML

Unit No	Unit Name	Prepared by		
		Name	Department	Signature
I				
II				
III				
IV				
V				

Verified by

Academic Coordinator

Head of the Department

Name :

Name :

Name :

Signature :

Signature :

Signature :

Date :

Date :

Date :

Note:

1. Verification to be done by senior faculty as allocated by the Department HOD.
2. Academic Coordinator and HOD of the department to which the faculty belongs must sign.
3. Verification for common subjects to be done by the Course Coordinator. In this case, Academic Coordinator and HOD of the department to which the course coordinator belongs must sign.

Knowledge Level as per Bloom's Taxonomy

K1- Remember; K2- Understand; K3- Apply; K4- Analyse; K5- Evaluate; K6- Create

K Level	Bloom's Taxonomy Action Verbs
K1	Choose, Define, Find, How, Label, List, Recall, Relate, Select, Show, Spell, What, When, Where, Which, Who, Why
K2	Classify, Compare, Contrast, Demonstrate, Explain, Extend, Illustrate, Infer, Interpret, Outline, Relate, Rephrase, Show, Summarize, Translate
K3	Apply, Build, Choose, Construct, Develop, Experiment with, Identify, Interview, Make use of, Model, Organize, Plan, Select, Solve, Utilize
K4	Analyze, Assume, Categorize, Classify, Compare, Conclusion, Contrast, Discover, Dissect, Distinguish, Divide, Examine, Function, Inference, Inspect
K5	Appraise, Assess, Award, Choose, Compare, Conclude, Criticize, Decide, Deduct, Defend, Determine, Disprove, Estimate, Evaluate, Explain, Importance, Influence, Interpret, Judge, Justify, Measure, Opinion, Perceive, Prioritize, Prove, Recommend
K6	Adapt, Build, Change, Choose, Combine, Compile, Compose, Construct, Create, Delete, Design, Develop, Discuss, Elaborate, Estimate, Formulate, Happen, Imagine, Improve, Invent, Makeup, Maximize, Minimize, Modify, Original, Originate, Plan, Predict, Propose, Solution, Solve, Suppose, Test, Theory

Unit I

BOOLEAN ALGEBRA AND LOGIC GATES

Course Objective – COB1: To understand the digital circuits using simplified Boolean Function

Course Outcome - CO1 : Design the digital circuits using simplified Boolean Functions.

Bloom's Taxonomy Level: K1,K2,K3

Part A

S.No.	Question	BTL*
1.	Write the names of basic logical operators. 1. AND 2. OR 3. NOT/INVERT	K1
2.	What are basic properties of Boolean algebra? The basic properties of Boolean algebra are commutative property, associative property and distributive property.	K1
3.	State the associative property of Boolean algebra. The associative property of Boolean algebra states that the OR ing of several variables results in the same regardless of the grouping of the variables. The associative property is stated as follows: $A+(B+C)=(A+B) +C$	K1
4.	State the commutative property of Boolean algebra. The commutative property states that the order in which the variables are ORed makes no difference. The commutative property is: $A+B=B+A$	K1
5.	State the distributive property of Boolean algebra. The distributive property states that AND ing several variables and ORing the result with a single variable is equivalent to OR ing the single variable with each of the several variables and then AND ing the sums. The distributive property is: $A+BC=(A+B)(A+C)$	K1
6.	State the absorption law of Boolean algebra. The absorption law of Boolean algebra is given by $X+XY=X$, $X(X+Y)=X$.	K1
7.	State DeMorgan's theorem. DeMorgan suggested two theorems that form important part of Boolean algebra. They are, 1) The complement of a product is equal to the sum of the complements. $(AB)' = A' +$	K1

	B' 2) The complement of a sum term is equal to the product of the complements. $(A+B)'=A'B'$	
8.	Reduce $A.A'C$ $A.A'C = 0.C$ [$A.A'=1$] $=0$	K1
9.	Reduce $A(A+B)$ $A(A+B) = AA+AB$ $=A(1+B)$ [$1+B=1$] $=A$.	K1
10.	Define duality property. Duality property states that every algebraic expression deducible from the postulates of Boolean algebra remains valid if the operators and identity elements are interchanged. If the dual of an algebraic expression is desired, we simply interchange OR and AND operators and replace 1's by 0's and 0's by 1's.	K1
11.	Find the complement of the functions $F1 = x'yz' + x'y'z$ and $F2 = x(y'z' + yz)$. By applying De-Morgan's theorem. $F1' = (x'yz' + x'y'z)' = (x'yz')'(x'y'z)' = (x+y'+z)(x+y+z')$ $F2' = [x(y'z' + yz)]' = x' + (y'z' + yz)'$ $= x' + (y'z')'(yz)' = x' + (y+z)(y'+z')$	K1
12.	Simplify the following expression $Y = (A + B) (A = C) (B + C)$ $= (AA + A C + AB + BC) (B + C)$ $= (AC + A B + B C) (B + C)$ $= A B C + A C C + A B B + A B C + B B C + B C C = A B C$	K1
13.	What are the methods adopted to reduce Boolean function? i) Karnaugh map ii) Tabular method or QuineMc-Cluskey method iii) Variable entered map technique.	K1
14.	State the limitations of karnaugh map. i) Generally, it is limited to six variable map (i.e) more than six variable involving expression are not reduced. ii) The map method is restricted in its capability since they are useful for simplifying only Boolean expression represented in standard form.	K1

15.	What is a karnaugh map? A karnaugh map or kmap is a pictorial form of truth table, in which the map diagram is made up of squares, with each squares representing one minterm of the function.	K1
16.	Find the minterms of the logical expression Y $=A'B'C'+A'B'C+A'BC+ABC'$ $Y=A'B'C'+A'B'C+A'BC+ABC'$ $=m_0+m_1+m_3+m_6 = m(0, 1, 3, 6)$	K1
17.	Write the maxterms corresponding to the logical expression $Y=(A+B+C')(A+B'+C)(A'+B'+C)$ $=(A+B+C')(A+B'+C)(A'+B'+C)$ $=M_1.M_3.M_6 = M(1,3,6)$	K4
18.	What are called don't care conditions? In some logic circuits certain input conditions never occur, therefore the corresponding output never appears. In such cases the output level is not defined, it can be either high or low. These output levels are indicated by 'X' or in the truth tables and are called don't care conditions or incompletely specified functions.	K1
19.	What is a Logic gate? Logic gates are the basic elements that make up a digital system. The electronic gate is a circuit that is able to operate on a number of binary inputs in order to perform a particular logical function.	K4
20.	Which gates are called as the universal gates? What are its advantages? The NAND and NOR gates are called as the universal gates. These gates are used to perform any type of logic application.	K1

Part B

S.No.	Question	BTL *
1.	Simplify (8) (i) $BC(AD+AD')+AB'$ (ii) $C(B+C)(A+B+C)$	K3
2.	Explain about different types of logic gates.(6)	K2
3.	Reduce the following function using k-map technique $f(A,B,C,D)=$ $\Sigma m(0,3,4,7,8,10,12,14)+d(2,6)$ (10)	K2
4.	Simplify the Boolean function using K-map. $F(A,B,C,D) = \Sigma m$ $(0,2,3,4,6,7,9,10,12,13,)$ (10)	K2
5.	Reduce the expression $F = \Sigma m(1,6,12,13,14) + d(2,4)$ is SOP and POS forms. (10)	K2

6.	Simplify the boolean function $F(A,B,C,D) = \sum m (1,3,6,7,8,9,10,12,14,15) + \sum d (11,13)$ (10)	K2
7.	Explain various theorems of Boolean algebra.(8)	K2
8.	Explain the various codes used in digital systems with an example.(6)	K2